Supplement Section A

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In the .Pdf output file you may find only the outputs generated by the analysis code, in order to keep the supplementary files as concise as possible. However the full code is available on https://github.com/Rezvani-S/Costly-Beauty-ShoeStudy.git

Part I: Stimuli acquisition Participant Demographics

Table A1.1: Stimuli acquisition Participant Characteristics

Table 1: Stimuli acquisition Participant Demographics (N = 60)

Variable	N or M	% or SD	Range
Demographics			
Sample size	60.00		
Age (years)	23.47	(3.62)	18-35
Height (cm)	167.93	(5)	156-180
Weight (kg)	59.80	(6.23)	48-76
BMI	21.18	(1.81)	18.04-25.59
Nationality			
Czech	48.00	80%	
Slovak	9.00	15%	
Other	3.00	5%	
Education			
Basic	1.00	1.7%	
Secondary school	35.00	58.3%	
Bachelor's degree	16.00	26.7%	
Master's degree	8.00	13.3%	
Sexual Orientation			
Heterosexual	42.00	70%	
Bisexual	14.00	23.3%	
Homosexual	2.00	3.3%	
Other/Unknown	2.00	3.3%	
Relationship Status			
Committed relationship	30.00	50%	
Single	20.00	33.3%	

Marriage/Partnership	3.00	5%
Non-committed	6.00	10%
Municipality		
Large City >100k	41.00	68.3%
Other	19.00	31.7%
Income		
Up to 15k CZK	39.00	65%
15-30k CZK	12.00	20%
30-50k CZK	8.00	13.3%

Table A1.2: Stimuli acquisition Missing Data

Table 2: Stimuli acquisition Missing Data Summary

Variable	Missing_Count	Missing_Percentage
income	1	1.67

Part II: Stimuli Acquisition — Participants, Procedure, and Equipment

Participants

Sixty women were recruited as stimulus models. Eligibility criteria were age 18–34 years, a BMI within the normal weight range (18.5–24.9) and an EU shoe size of 37–39. Sizes 37–39 represent the most common women's shoe sizes (Sherman, 2023), the age range captures young adults during their peak mating age (Buunk et al., 2001), and the BMI restriction was implemented to enable the hypothesised effects of high heels on body posture (Lewis et al., 2017). Demographic data were collected via an online questionnaire that did not record participant codes and are therefore reported for all 60 recruited participants (Table A1; age: M = 23.47, SD = 3.62, range 18–35 years; BMI: M = 21.19, SD = 1.81, range 18.0–25.6). Two participants were unable to complete the heeled-walking trials on the treadmill and were excluded, yielding 58 recorded models. For the rating study, models were randomly allocated to 11 rating blocks of five models (55 models), leaving three recorded models unassigned; owing to an allocation error, one block contained four models, so the final stimulus set comprised 54 models. Participant recruitment occurred in January 2024 via social media platforms and in-person flyer distribution across several faculties of Charles University. Data collection was conducted during February and March 2024 at the Faculty of Physical Education and Sport, Charles University. Participants received compensation of 400 CZK upon completion.

Procedure

Data collection occurred in a laboratory setting. Upon arrival, participants provided informed consent, completed sociodemographic and standardised questionnaires via Google Forms, and changed into provided standardised attire (black leggings, a stretch black tank top, and nude-coloured one-time-use socks). Hair was secured away from the face and accessories were removed to maintain a standardised appearance. Before stimuli capture, participants selected the appropriate shoe size from available options (sizes 37, 38, and 39) for each shoe type.

A familiarisation period allowed participants to walk on the treadmill and adjust to each shoe type under safe precautions, during which they practised maintaining a natural walking gait while looking ahead, not holding the treadmill handles, and walking in the middle of the belt. Treadmill sidebars were folded during image and video capture to avoid visual interference. Footwear items were worn in a randomised order determined by a computer-generated random sequence.

For each condition, participants initially stood at the treadmill midpoint — feet hip-distance apart, toes pointing straight ahead, equal weight on both feet, back straight, hands resting at the sides with palms open, and gaze directed into the camera lens — for frontal and right-side profile image capture. For video recording, participants began walking while a research assistant gradually increased treadmill speed to a constant rate of 1.6 km/h in heel and control conditions and 2.4 km/h in flat conditions. These speeds approximated comfortable gait velocity for each footwear type as established through pre-data-collection procedural dry runs. Recording commenced after verbal confirmation of comfortable walking. Thirty seconds of uninterrupted gait footage per condition was obtained, with re-recording performed where walking requirements were not met. Each participant's session lasted approximately one hour.

Equipment and Setup

Stimuli were acquired in a controlled laboratory environment with external light fully blocked to eliminate ambient variation. The scene was illuminated with an Amaran 300c LED light equipped with a Fomei 150 cm parabolic softbox with a front diffuser panel (intensity: 100%; daylight colour balance: 5600 K, Hue: 180°, Saturation: 0%), positioned at the 2 m mark directly above the camera axis with a 10° downward tilt. Correct exposure was verified using a digital light meter (Sekonic L-308S).

A Canon EOS 250D APS-C digital camera fitted with a Canon EF-S 18–55 mm f/4–5.6 IS STM lens was used for both still image and video acquisition, mounted on a tripod in portrait orientation at a fixed subject

distance of 2 m (measured from the camera sensor plane). For video acquisition, the lens was set to a focal length of 18 mm (approximately 29 mm full-frame equivalent), with aperture f/5.6, shutter speed 1/50 s, and ISO 400. Videos were recorded at 25 fps in 4K resolution (3840×2160 px) in MP4 format (H.264/AVC, IPB compression, average bitrate 119 Mbps), with rec.709 colour primaries at full range; in-camera white balance was set to Daylight (5200 K) and colour profile to Standard. Full-body photographs were acquired with identical camera settings in RAW format (CR3, 3840×2160 px native sensor output cropped to 4020×6024 px).

The treadmill (Urevo Strol 3) was painted matte white to maximise contrast with the stimuli, and a white canvas backdrop was positioned behind the subject. Tripod positions were marked on the floor to ensure consistent camera placement across all sessions. Colour and white balance calibration were performed at the beginning of each session using a reference photograph of a ColorChecker Passport Photo 2 calibration card, with the 18% grey patch serving as the white balance reference.

Stimuli Postprocessing

Image processing. Image stimuli were processed using Adobe Photoshop Lightroom Classic 13.1. Full-body photographs were standardised using a grid-based cropping procedure, with colour and white balance calibrated from the session-specific reference photographs. The lower boundary of the crop frame was aligned to the treadmill edge to exclude extraneous background elements; horizontal centering was achieved by aligning the image midline with the longitudinal body axis; and vertical framing was standardised by positioning the upper third grid line at shoulder height. Due to natural variation in participant stature, minor variation in crop dimensions was unavoidable; however, final cropped images (approximately 1565×3184 pixels) exceeded the 1080×1920-pixel output resolution of the projection system, preserving stimulus clarity. Facial features were obscured using a circular mask with a 450-pixel radius and a 35% mosaic blur. Processed images were exported in JPEG format at maximum quality (100%) with sRGB colour space encoding.

Video processing. Video stimuli were processed using DaVinci Resolve 20. Videos were cropped to portrait orientation (1080×1920 px) to match the projection system's output resolution, with the lower boundary fixed at the treadmill edge. To maintain consistent horizontal centering of moving subjects, a circular power window mask applied over each participant's head region was tracked across all frames using DaVinci Resolve's integrated object tracking function. Facial anonymisation was achieved via a Mosaic Blur OpenFX effect applied to the tracked region (pixel frequency: 50.0; square cell shape; anti-aliasing: 1.00). Final video segments consisted of 10 seconds of uninterrupted walking trimmed from the 30-second recorded footage.

Of the 60 female participants in the stimuli acquisition phase, two were excluded from stimuli processing as it proved impossible to obtain 30 seconds of uninterrupted gait footage in heel conditions due to substantial walking difficulty; all received full compensation. From the remaining 58 processed stimulus sets, 54 were utilised in the rating phase (one block included 10 models by a mistake that was found after data collection), with three unselected as an incidental byproduct of the random assignment of models to rating blocks. From all body images and videos, 11 rating blocks were created by randomly assigning 5 models to each block; each block contained 25 frontal full-body images and 25 gait videos across all five conditions (control, economy flat, economy heel, luxury flat, and luxury heel).

Part III: Rating session Participant Demographics

Table A3.1: Rating session Participant Characteristics

Table 3: Rating session Participant Demographics (N = 279)

Variable	N or M	% or SD	Range
Demographics			
Sample size	279.00		
Age (years)	22.66	(3.71)	18-40
Sex			
Female	177.00	63.4%	
Male	102.00	36.6%	
Nationality			
Czech	237.00	84.9%	
Slovak	25.00	9%	
Other	17.00	6.1%	
Education			
Elementary school	7.00	2.5%	
High school	153.00	54.8%	
Bachelor's degree	77.00	27.6%	
Master's degree	37.00	13.3%	
Ph.D. degree	3.00	1.1%	
Other	2.00	0.7%	
Sexual Orientation			
Exclusively heterosexual	143.00	51.3%	
Predominantly heterosexual	75.00	26.9%	
Bisexual	30.00	10.8%	
Rather heterosexual	17.00	6.1%	
Exclusively homosexual	7.00	2.5%	
Rather homosexual	5.00	1.8%	
Predominantly homosexual	2.00	0.7%	
Partnership Status			
Serious relationship	126.00	45.2%	
Single (no partner)	126.00	45.2%	
Non-binding relationship	16.00	5.7%	
Marriage/registered partnership	8.00	2.9%	
Other	2.00	0.7%	
Residence			
Large city (>100k)	198.00	71%	
Other	81.00	29%	
Income			
Up to 15k CZK	206.00	73.8%	
15-30k CZK	48.00	17.2%	
30-50k CZK	18.00	6.5%	
Over 50k CZK	6.00	2.2%	
Survey Completion			
Survey Duration (minutes)	43.41	(29.74)	

Table A3.2: Rating session Missing Data

Table 4: Rating session Missing Data Summary

Variable	Missing_Count	Missing_Percentage
partnershipstatus	1	0.36
monthlyincome	1	0.36
partnershipstatus_clean	1	0.36

Part IV: Effect Size Interpretation Guidelines

Effect Size Interpretation

Effect size interpretations in this study follow empirically-derived guidelines from individual differences research rather than Cohen's (1988) conventional benchmarks. Cohen's original thresholds ($d = 0.20/0.50/0.80$ for small/medium/large) were based on subjective impression rather than empirical data and have been shown to overestimate typical effect sizes in psychological research (Lovakov & Agadullina, 2021).

We adopted the guidelines proposed by Gignac and Szodorai (2016), who analyzed 708 effect sizes from individual differences research and recommended interpreting correlations of $r = 0.10$, 0.20, and 0.30 as small, medium, and large, respectively. Converting these correlation benchmarks to Cohen's d using the formula $d = 2r/\sqrt{(1\text{-}r^2)}$, we interpret effect sizes as follows: $d = 0.22$-0.39 as small, $d = 0.40$-0.59 as medium, and d 0.60 as large. Effect sizes below $d = 0.22$ are characterized as "very small."

This choice of guidelines is particularly appropriate given that our study examines individual differences in perceptual ratings. Gignac and Szodorai's (2016) benchmarks reflect the magnitude of effects commonly observed when measuring stable individual differences in psychological constructs, which aligns with our investigation of how individuals differ in their perceptions of attractiveness and wealth based on appearance cues.

Additionally, as our study was pre-registered, it is important to contextualize our findings within the broader literature on publication bias and effect size inflation. Schäfer and Schwarz (2019) demonstrated that studies without pre-registration show substantially inflated effect sizes (median $r = 0.36$) compared to pre-registered studies (median $r = 0.16$), suggesting that publication bias approximately doubles observed effects. Our pre-registered design likely yields more accurate estimates, and the relatively small effects we observed should be interpreted as more reliable indicators of true effect magnitudes.

For completeness, we also provide approximate conversions to partial eta squared (^2_p) using the formula ^2_p $d^2/(d^2 + 4)$ (Cohen, 1988; Lakens, 2013). This allows comparison with ANOVA-based effect sizes commonly reported in experimental psychology.

Table 5: Effect Size Interpretation Guidelines for Individual Differences Research

Effect Size (d)	Interpretation	Equivalent r	Partial ^{2}	Basis
< 0.22	Very small	< 0.11	< 0.012	Below Gignac & Szodorai (2016) small threshold
0.22 - 0.39	Small	0.11 - 0.19	0.012 - 0.037	Gignac & Szodorai (2016) small effect
0.40 - 0.59	Medium	0.20 - 0.29	0.038 - 0.080	Gignac & Szodorai (2016) medium effect
0.60	Large	0.30	0.083	Gignac & Szodorai (2016) large effect

Part V: Questionnaire Instruments (English Translations)

The following instruments were administered in Czech. English translations are provided below for reporting purposes

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A5.1 Sociodemographic Questionnaire

1. What gender do you identify with?

- Male
- Female
- Other (please specify): _____
- Prefer not to say

2. What is your biological sex as stated on your ID?

- Male
- Female

3. How old are you?

(Dropdown: 15–99)

4. What is your nationality?

- Czech
- Slovak
- Other (please specify): _____

5. What is your sexual orientation?

- Exclusively heterosexual
- Predominantly heterosexual
- Mostly heterosexual
- Bisexual
- Mostly homosexual
- Predominantly homosexual
- Exclusively homosexual

6. What is your relationship status? *(Select all that apply)*

- Single (without a partner)
- Casual relationship
- Committed relationship
- Marriage / registered partnership
- Other: _____

7. What is your highest level of education completed?

- Primary school
- Secondary school with vocational certificate
- Secondary school with graduation exam (Maturita)
- University – Bachelor's degree
- University – Master's degree
- University – Doctoral degree (Ph.D.)
- Other: _____

8. What is the size of the municipality where you currently live?

- Village up to 1,000 inhabitants
- Municipality with 1,001–5,000 inhabitants
- Municipality with 5,001–20,000 inhabitants
- Town with 20,001–50,000 inhabitants
- Town with 50,001–100,000 inhabitants
- City with over 100,000 inhabitants

9. What is your average monthly income?

- Up to 15,000 CZK/month
- 15,000–30,000 CZK/month
- 30,000–50,000 CZK/month
- Over 50,000 CZK/month

A5.2 Subjective Fashion Knowledge Scale

Flynn & Goldsmith (1999); 5-point Likert scale: 1 = Strongly disagree — 5 = Strongly agree

1. I know quite a lot about fashion clothing.

2. I can assess the quality of a piece of clothing.

3. I think I know enough about fashion clothing to feel confident when buying.

4. I don't feel very knowledgeable about fashion. *(reverse-scored)*

5. Among my friends, I am one of the "experts" on fashion clothing.

6. Compared to most other people, I know less about fashion clothing. *(reverse-scored)*

7. I have heard about most new fashion trends.

8. When it comes to fashion, I don't know very much. *(reverse-scored)*

9. I can tell whether a fashion item is worth the price or not.

A5.3 Fashion Brand Knowledge Scale

Purpose-designed instrument. Participants rated each brand on three dimensions using a 0–7 scale, and selected applicable brand attributes.

Per-brand items *(administered identically for each of the 10 brands listed below)*:

Item	Scale
How well do you know this brand?	0 (not at all) – 7 (very well)
How accurately can you recognise a product of this brand?	0 (not at all) – 7 (very well)
How would you categorise this brand?	1 (budget) – 7 (luxury)

Attribute checklist *(select all that apply)*:

- Exclusive
- High quality
- Prestigious
- Expensive
- Authentic
- Rich brand heritage
- Aesthetic appeal

Brands assessed:

Economy brands: H&M, Zara, Primark, Shein, Sinsay

Luxury brands: Yves Saint Laurent (YSL), Chanel, Gucci, Dior, Givenchy

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